SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

ICT330

Examination - January Semester 2025

Database Management Systems

Monday, 14 April 2025 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **4** questions and comprises **6** printed pages (including cover page).
2. You must answer **4** questions.
3. All answers must be written in the answer book.
4. This is **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1

A ride hailing platform allows registered customers to order a ride at a fare determined by the platform. A customer specifies a pickup point and a drop off point, and at most one additional stop between the pickup and drop off points.

The address of each location, whether a pickup point, an additional stop if required, or a drop off point, is recorded for each ride, in addition to the time of booking, the actual time of pickup and the completion time of the journey, the number of passengers and the fare.

A location is identified by a postal code. Locations can be saved under the customer's favourite locations.

A registered driver who is near the pickup point of the order can decide whether to accept the order. Once the platform confirms a driver to an order, no other driver can accept the order.

A registered driver has a driver id, name, date that the driver joins the platform and one car which must be registered with and approved by the platform. The car registration number, make, model and car colour are recorded. A car can be shared by multiple driver.

Without adding any attribute/identifier not described in the question, construct a conceptual entity relationship (ER) model from the statements of requirements to represent the data model, showing the data requirements in one diagram.

State your assumption(s) for only data requirements that are not specified.

Question 1a

Entities, with entities name, relevant attributes and identifier;

(10 marks)

Question 1b

Relationships, with maximum and minimum cardinalities and relationship name.

(10 marks)

Question 2

You are given the table:
PatientLabTest(patientId, patientName, patientDOB, labTestCode, labTestName, labTestRequirement, testDate, labTestResult)

One or more lab test can be performed for a patient on a given test date. A lab test is identified by a *labTestCode* and a patient is identified by *patientId*. A lab test may have one or more requirements e.g., fasting, stop certain medication, avoid strenuous exercise etc.

Question 2a

Formulate and list the functional and multi-valued dependencies (MVD). Include at least **ONE (1)** MVD that is subsumed and at least **ONE (1)** MVD that is not subsumed.

Do not introduce any new columns not in the given table. State your assumption(s) for only data requirements that are not specified.

(5 marks)

Question 2b

Draw dependency graph and propose candidate key(s) for the table. Show composite keys in brackets.

(3 marks)

Question 2c(i)

For each MVD in answer to Q2(a), explain why it is subsumed or not subsumed.

(2 marks)

Question 2c(ii)

Normalise the table to Boyce Codd Normal Form (BCNF) and Fourth Normal Form (4NF) by applying the **BCNF approach** covered in the course text.

Show how you apply the steps to arrive at the tables, explain and discuss whether the structure of each derived table at each iteration is already in BCNF and 4NF (if applicable).

(8 marks)

Question 2d

Assemble your BCNF and 4NF tables and constraints clearly by giving each table and the referential integrity constraint(s), if any, in the forms described below:
Tables in this form:

TABLE-NAME(col-name$_1$, col-name$_2$, ..., col-name$_n$)

- indicate (PK) against the primary key and
- indicate (FK) against the foreign keys, if any.

Referential integrity constraint(s), if any, **together with the child table** in this form:

TABLE-NAME$_x$.col-name$_{x1}$ ...col-name$_{xn}$ must exist in

TABLE-NAME$_y$.col-name$_{y1}$...col-name$_{yn}$

(7 marks)

Question 3

Question 3a(i)
Transform the ERD in Figure Q3(a)(i) into a logical model.

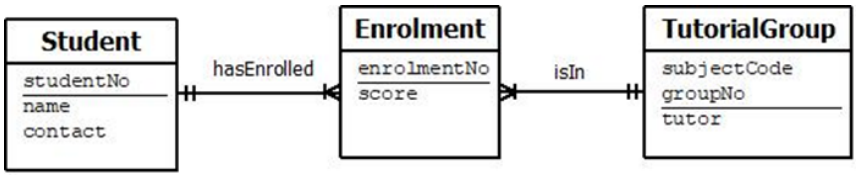

Figure Q3(a)(i)

Show the tables in the logical model in the form:

TABLE-NAME(col-name$_1$, col-name$_2$, ..., col-name$_n$)

- indicate (PK) against the primary key,
- indicate (FK) against the foreign keys, if any, including integrity constraints in the form:

  TABLE-NAME$_x$.col-name$_{x1}$ ...col-name$_{xn}$ must exist in

  TABLE-NAME$_y$.col-name$_{y1}$...col-name$_{yn}$

(4 marks)

Question 3a(ii)

Refer to the description of the columns for the column definition, and then code the appropriate SQL statements to **implement ONLY** the table derived for the entity Enrolment in Figure Q3(a)(i).

All columns are required except for the column score.

studentNo	Consists of one alphabet followed by 8 digits.
name	At most 40 characters in length
contact	Consists of 8 digits
enrolmentNo	An integer, auto generated with starting value 1 and increment 1
score	An integer between 0 and 100
subjectCode	Consists of three alphabets followed by 3 digits.
groupNo	An integer between 1 and 20
tutor	At most 40 characters in length

Constraints for the relationships in Figure Q3(a)(i):

hasEnrolled	delete and update are cascaded
isIn	delete disallowed and update cascaded

(6 marks)

Question 3b

Formulate SQL queries to perform the following tasks:

Question 3b(i)

List the subjectCode, groupNo and tutor of tutorial groups for subject codes that start with 'ICT'. Sort the result in ascending order of subjectCode and groupNo.

(4 marks)

Question 3b(ii)

List the studentNo and name of students who are currently enrolled in both ICT133 and ICT162.

(6 marks)

Question 3b(iii)

List the subjectCode, groupNo and tutor of tutorial groups where there are at least 2 students whose scores are recorded as null.

(5 marks)

Question 3c

Write a trigger to perform the following actions when a student is placed into a tutorial group:

- The student must not already be in a tutorial group for the same subject code.
- The student must not be in more than 4 tutorial groups.

State any assumptions you made in implementing the trigger.

(10 marks)

Question 4

Question 4a

Consider the database for View Ridge Gallery with the tables: Customer, Trans, Work and Artist. Suppose two customers are viewing the available works in the gallery online, and both wish to purchase the same piece of artwork.
Propose an isolation level where lost update will not happen. Explain why the proposed isolation level cannot cause lost update to happen.

(6 marks)

Question 4b

Describe and demonstrate **ONE (1)** permission that should be given to students using the database for Question 3, based on the ERD shown in Figure Q3(a)(i). Describe **ONE (1)** permission that should not be given to students. Specify the permissions in terms of these actions: select, delete, insert and update on user tables only.

(4 marks)

Question 4c

Given Figure 4(c), explain what the database recovery procedure will do for the transactions T1 to T3. Explain also how the transaction log help ensure the ACID properties of T1 to T3.

Record Pointer	Transaction ID	Reverse Pointer	Forward Pointer	Time	Type of Operation	Table	Row ID	Before Image	After Image
1	T1	0	2	11:01:00	Start				
2	T1	1	3	11:01:36	Delete	Table1	2	2, …	
3	T1	2	0	11:02:55	Delete	Table2	14	14, 2, …	
4	T2	0	5	11:03:01	Start				
5	T2	4	6	11:03:24	Update	Table3	15	15, 3, …	15,5, …
6	T2	5	7	11:04:41	Update	Table4	16	16, 4, …	16,6, …
7	T2	6	10	11:05:05	Delete	Table1	3	2, …	
Checkpoint									
8	T3	0	9	11:23:23	Start				
9	T3	8	12	11:25:00	Insert	Table1	14		14,8, ...
10	T2	7	11	11:26:02	Insert	Table2	23		23,3, …
11	T2	10	0	11:27:25	Rollback				
12	T3	9	0	11:28:33	Commit				
Crash!									

Figure 4(c)

(10 marks)

----- END OF PAPER -----